

APPENDIX C-1 COST MODELS AND COST METHODOLOGY

1. Cost Models & Costing Methodology

The cost models are the underlying methodology and equations that relate the final cost of a part to the different operations and goods used in that part. These descriptions are accurate at the time of the rules publications. The models may be expanded as necessary based on evolving requirements.

2. Raw materials

- 2.1. Raw materials refer to the material stocks used to produce parts from scratch, such as billet steel for machining or aluminum ingot for casting. Bar, sheet and tube stock are purchased using raw material costs. The raw material purchased must include machining allowance. Standard allowances are given in 4.1 and must be used
- 2.2. Gross weight will refer to the weight of the raw material, including all machining stock
- 2.3. Net weight will refer to the weight of the finish machined part
- 2.4. Material costs are based on part gross weight. For example a steel hub is machined from solid bar. The interior is removed by boring. The cost of the bar must include this interior material. Raw materials are normally cost by volume. A cost by weight is also given using an official density listed in the tables. Any parts that are weighed at competition to confirm cost will use the official density in calculating cost.

3. Assembly Labor

The assembly labor model is based on the following parameters:

- 3.1. Mass – The mass of the part influences the time it takes the operator to assemble the part to the assembly or vehicle. Light parts can be installed with one hand. Heavier parts require two hands and the heaviest parts need a lift assist apparatus. These factors are accounted for by selecting the appropriate entry from the process labor tables. The actual part mass must be equal to or less than the value selected. For example a 300g part would have an assembly labor category of 1 kg.
- 3.2. Interfaces – The more interfaces a part has with the surrounding parts the longer it takes to assemble. Parts designed for minimal constraint are the easiest and cheapest to assemble.
- 3.3. Fit type – The ease with which a part can be assembled is described by the fit. There are three categories of fits:
 - 3.3.1. Loose – the part assembles with no force. Examples include a quick release steering

wheel onto the steering shaft and a bracket bolted to a monocoque.

- 3.3.2. Line on line – the part is designed to have a close fit to the surrounding parts and some build-up of force is required to get the part started. Examples include a rod end inserted between two tabs in double shear and a splined axle shaft into the differential gear.
- 3.3.3. Interference – significant force is required to insert the part and mechanical assistance may be necessary. Examples include a rubber hose onto a barbed fitting and a ball bearing into a bore.

4. Machining

Costs for machining operations are based on the volume of material removed. The actual machine used, whether mill, lathe or otherwise, is the same unless a specific line item is included for that machine, such as gear hob.

- 4.1. *All processes require a minimum of 1mm (0.040 inches) of machining stock to be removed from each surface of the part with machining.*
- 4.2. *The process multiplier for the material must also be used to calculate the total process cost of the operation. If a process multiplier is required it will be listed in the processes table in the column labelled 'Multiplier Type Used'. If the column is blank for a process none is required.*
- 4.3. *When costing the raw materials that go into making machined parts the machine stock must be included in the purchased material mass, even though this material is machined away to produce the final part. This represents the cost of the purchased material. For example, an upright bore is machined into a piece of billet aluminium. The interior material that is milled away must be included in the billet mass and hence cost. The same feature machined into a casting need only include 1mm of machine stock of the machined away material*
- 4.4. *Machining requires labor operations to account for the time it takes an operator to fixture the part onto the machine. Every machined part requires at least a 'Machining Setup, Install and Remove' operation. This is the time it takes to pick up the work piece, fixture on the machine, and remove it when the machining is complete. For a part that requires an intermediate change in position, such as to machine the back of the part which would not be accessible in a single fixturing setup, the labor step of 'Machining Setup, Change' is also required. For example, an upright that requires three different orientations on a mill to fully machine would require two of the 'Machining Setup, Change' and the 'Machining Setup, Install and Remove' labor operations.*
- 4.5. *In certain cases, it is possible to fixture a work piece of raw material and machine more than one part out of it. For example, a self-feeding lathe could machine 10 suspension inserts out of a single piece of bar stock. In this case the quantity of the 'Machining Setup, Install and Remove' may be set to 0.1. This represents the 10 parts that can be machined*

per setup. This assumption should clearly be noted in the Cost Report along with enough details for the Cost Judges to verify the part geometry is appropriate for the machine being used.

5. Tooling & Fixturing

Tooling is necessary when certain processes are used. These can be identified in the tables because the tooling required will be indicated. Sometimes several types of tooling are available for the same process. Each has a description and an associated process with which it can be used. If a process has more than one tooling type associated with it the team must use the tooling that is closest to the actual tooling used in their prototype vehicle construction. Most tooling costs are independent of part shape, the assumption being that tooling for smaller parts will be built with multiple cavities to create optimal cost effectiveness.

- 5.1. After calculating the total tooling cost for a part the cost must be divided by the Production Volume Factor (PVF) before being included in the Cost Report. The PVF represents the ability of the tooling to produce parts in volume production.

Production Volume Factor (PVF)

All parts not otherwise listed: 3000

Composite Monocoque (composite tub): 120

The following equation is used to calculate the tooling cost to be included for each part:

$$\text{Part Tooling Cost} = \frac{\text{Table Tooling Cost}}{\text{PVF} * \text{Number of Parts Using Exact Tooling}}$$

The tooling cost should be included with the appropriate part on the BOM. Tooling is not a separate section.

- 5.2. For example cast aluminium upright uses a 2-piece sand core package. Total table price is ` 325000 + ` 325000 = ` 650000. The team has designed the casting to be used for both the left and right hand rear corners. Calculating the Part Tooling Cost gives:

$$\text{Part Tooling Cost} = \frac{\text{` 650000}}{3000 * 2} = \text{` 108.55 per upright}$$

The ` 108.55 must be included as a line item on the Costed Bill of Material for each Upright.

- 5.3. Another example is a team is manufacturing a composite monocoque. The tube is constructed by building the top and bottom separately and bonding it together. Both the top and bottom use a two piece composite tool and the cost of all four tools is ` 29,25,000. The PVF for tube tooling is only 120 because of the amount of time required to construct each tube so the tube Part Tooling Cost is:

$$\text{Part Tooling Cost} = \frac{\text{₹} 29,25,000.}{120 * 1} = \text{₹} 24,375 \text{ per tube}$$

6. Fastener Installation

The cost to tighten or loosen fasteners is based on the tool (or motion) needed to turn it, the diameter, length and whether the fastener requires a secondary tool for reacting the torque (such as a wrench on a nut).

- 6.1. Hand – When no tool is necessary for tightening, such as quick release fasteners or hand tightened nuts, the hand should be selected. Loose operations are those accomplished by using the fingers of the hand. If the entire hand is moving to rotate the fastener the tight category should be used.
- 6.2. Screwdriver – A tool that can be held in the hand and turned with the wrist. Any type of bit can be fitted such as straight, Philips, Torx, etc.
- 6.3. Wrench – An open-ended or box wrench or similar tool requiring motion of the hand. After a turn the wrench may have to be removed and repositioned for the next turn.
- 6.4. Ratchet – A tool with internal clutch that allows the hand to be moved and returned to the starting position without removal of the tool. Compatible with any bolt head style such as 6-point hex, 12-point hex, Torx or other.
- 6.5. Power Tool – An electric, pneumatic or other power assisted tool for running down fasteners. To qualify for power tool use a fastener must meet the following requirements:
 - 6.5.1. A socket of the size needed to drive the fastener must fit in the fully secured position
 - 6.5.2. One power tool with minimum dimensions given in Appendix C-4 must fit onto the socket
 - 6.5.3. An extension may be used to fit the power tool but it may not exceed 0.35m in length
- 6.6. Reaction Tool - In the case where the fastener is not being attached into the part but requires a nut or other separate threaded piece then a reaction tool will be required. This will appear as a separate line item and should appear whenever a nut is used on a bolt.

7. Composites

Composite manufacture is defined by following steps:

- 7.1. Laminate – Used to build the laminate one (1) ply at a time. A ply is a single layer of the laminate consisting of a single sheet of material, regardless of material or thickness. A ply may consist of woven carbon, unidirectional glass, adhesive film or honeycomb core, for example.
- 7.2. Curing Operations – Used to take a laminate and convert it to a finished composite

structure. All curing operations include vacuum bagging, peel ply, breather cloth and other consumable materials and labor. Costs also include part removal from the mold.

- 7.2.1. Room temperature cure – used for room temperature curing resin systems.
- 7.2.2. Oven cure – used for higher temperature cure cycles for composites. Limited to one (1) atmosphere of external pressure.
- 7.2.3. Autoclave cure – used for high temperature and pressure composites curing.
- 7.3. Curing operations require tooling. Tooling must reflect the type of tooling actually used (composite, aluminum, steel, etc).
- 7.4. If hybrid weaves are used the cost can reflect the ratio of the materials in the ply. For example a 50% carbon fiber, 50% glass woven ply may use the average cost of the carbon and glass materials. If the actual fiber ratio is not used then the cost of the ply must be the cost of the highest cost material present.
- 7.5. *When costing composite materials the total mass of the part in the Cost Report must match the actual mass of the part as presented on the vehicle for Cost Judging. The composite material, whether carbon fiber or other must be the cost of both the fiber and resin together. This is true for both prepreg and dry fiber systems and is further stated in the Materials Table. The mass of each ply can be adjusted to make the finish part mass match the Cost Report. Parts can be weighed during judging and the Cost Report mass must be equal to or greater than the actual mass of the part, even including clear coat, paint and other finishes. Note: The paint mass is being included to avoid debates in Cost Judging about how much weight the paint (or clear coat) has added. The cost of the paint and paint application need not be included if it is solely for cosmetic purposes but the mass of paint must be included in the composite cost.*

8. Electronics and Wiring

The wiring harness is cost as a number of connectors of a certain style, each interconnected by a number of wires of a certain type. The electrical system is composed of three wiring types.

- 8.1. Signals - Inputs to the control system such as wheel speed, mass airflow or the position of a driver toggle switch.
- 8.2. Controls - Control system outputs. These can be digital signals, pulse width modulated or voltage outputs.
- 8.3. Power - Wires carrying current for vehicle distribution or actuators. These include vehicle power from the battery, engine starter, solenoids, motors etc.